

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

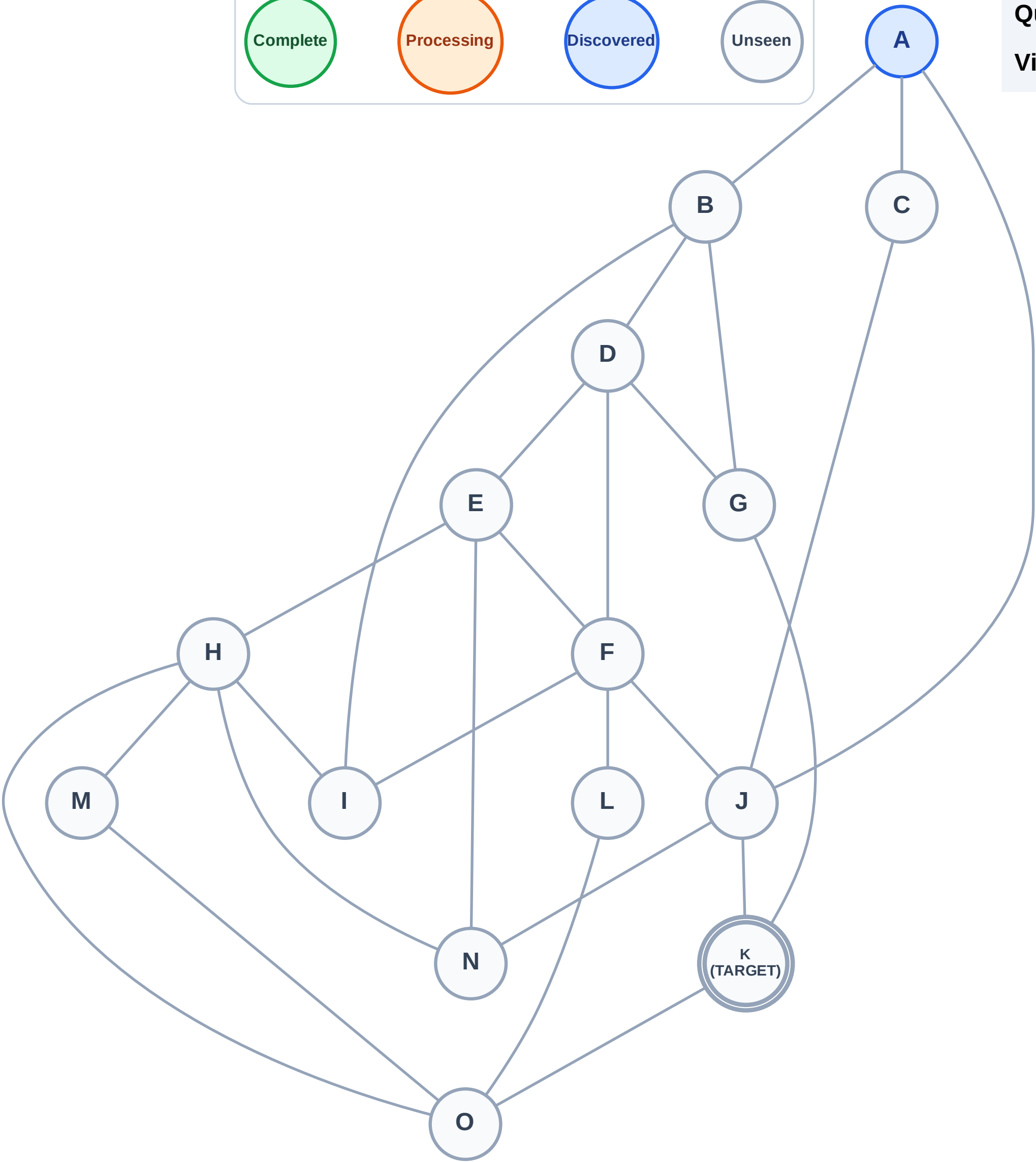
Complete

Processing

Discovered

Unseen

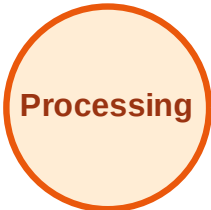
Queue: A
Visited: (none)



BFS Traversal

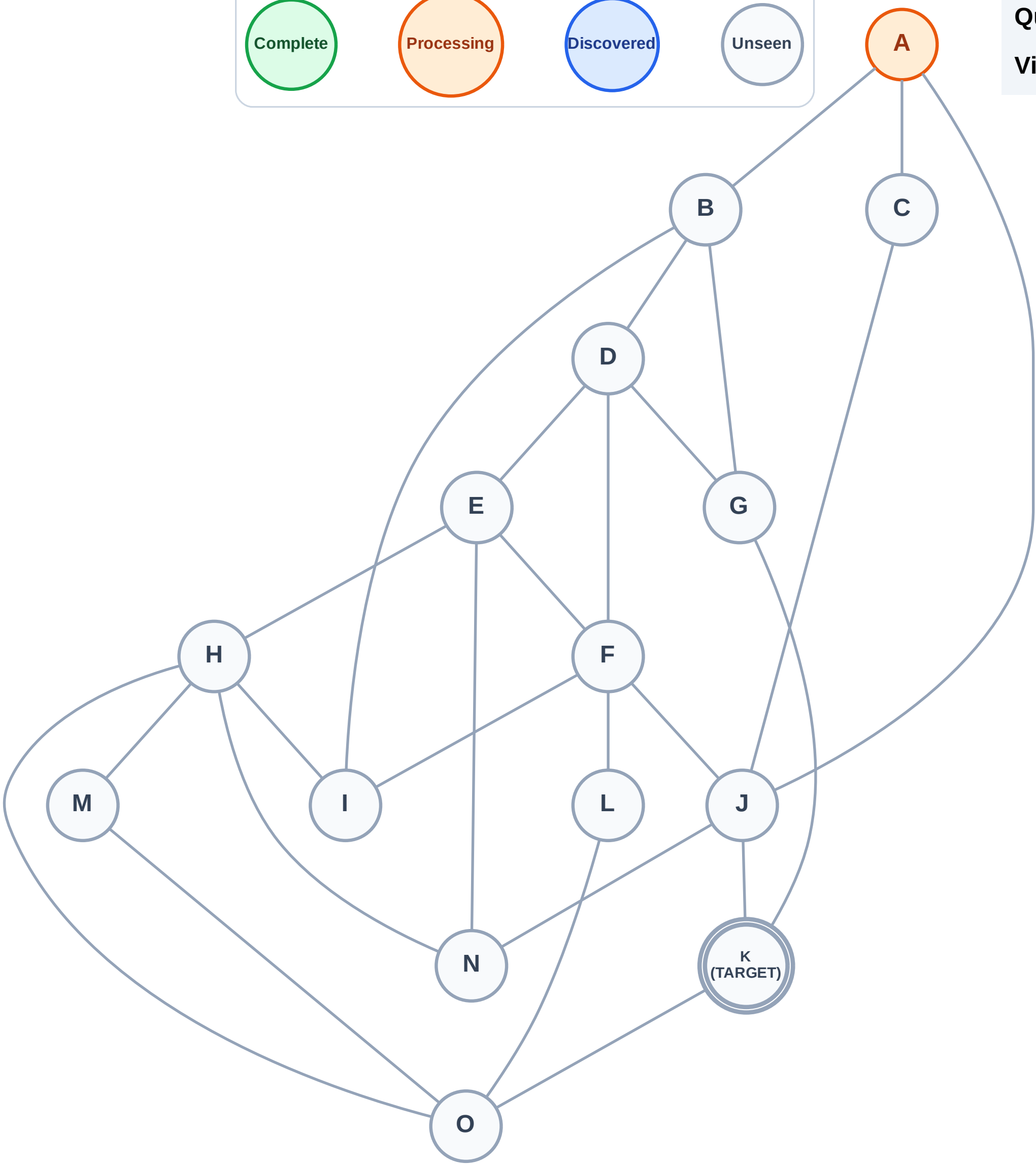
Step 2: Process node A

Legend



Queue: (empty)

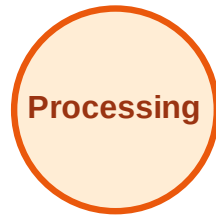
Visited: (none)



BFS Traversal

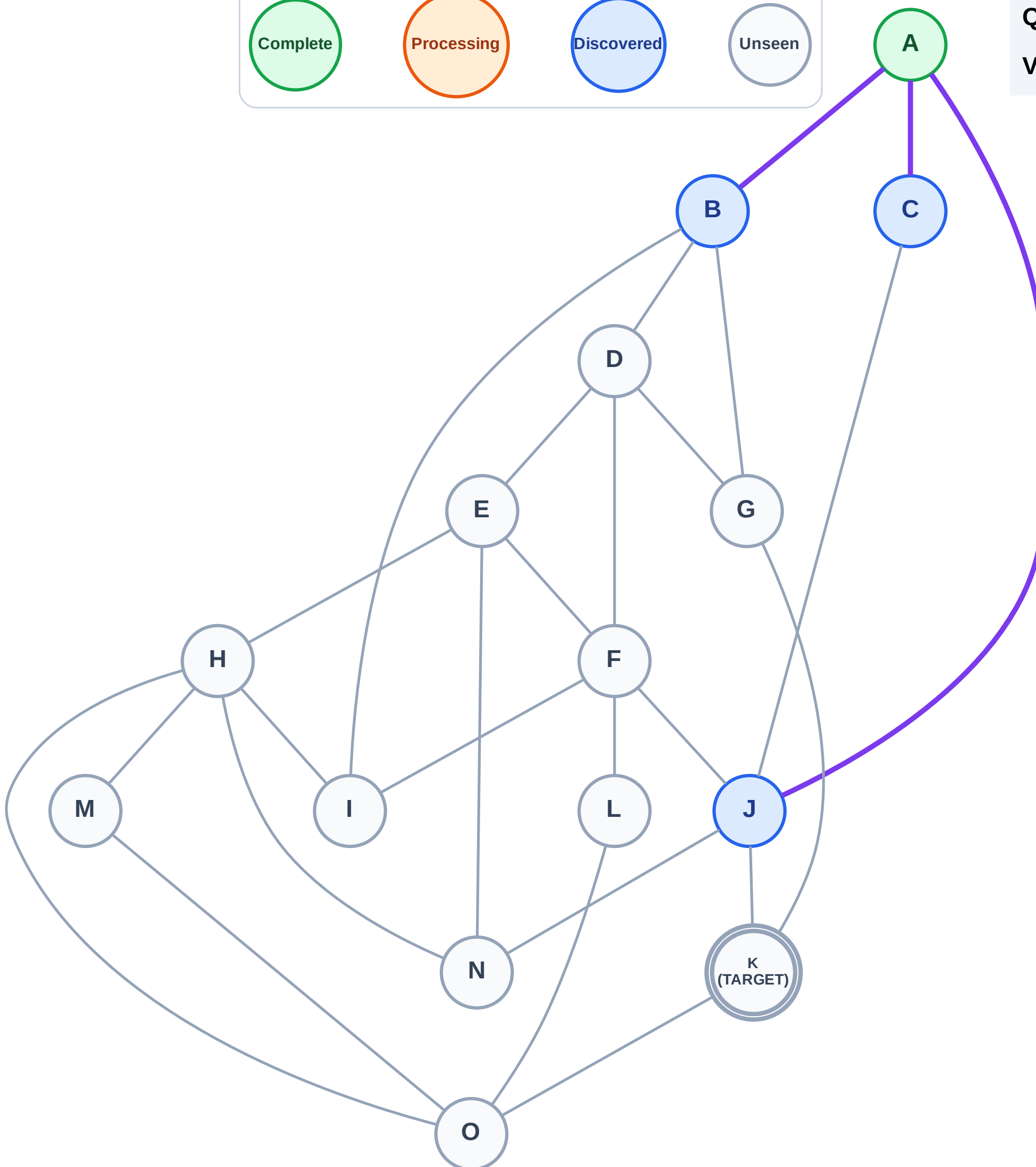
Step 3: Discover B, C, J from A

Legend



Queue: B → C → J

Visited: A



BFS Traversal

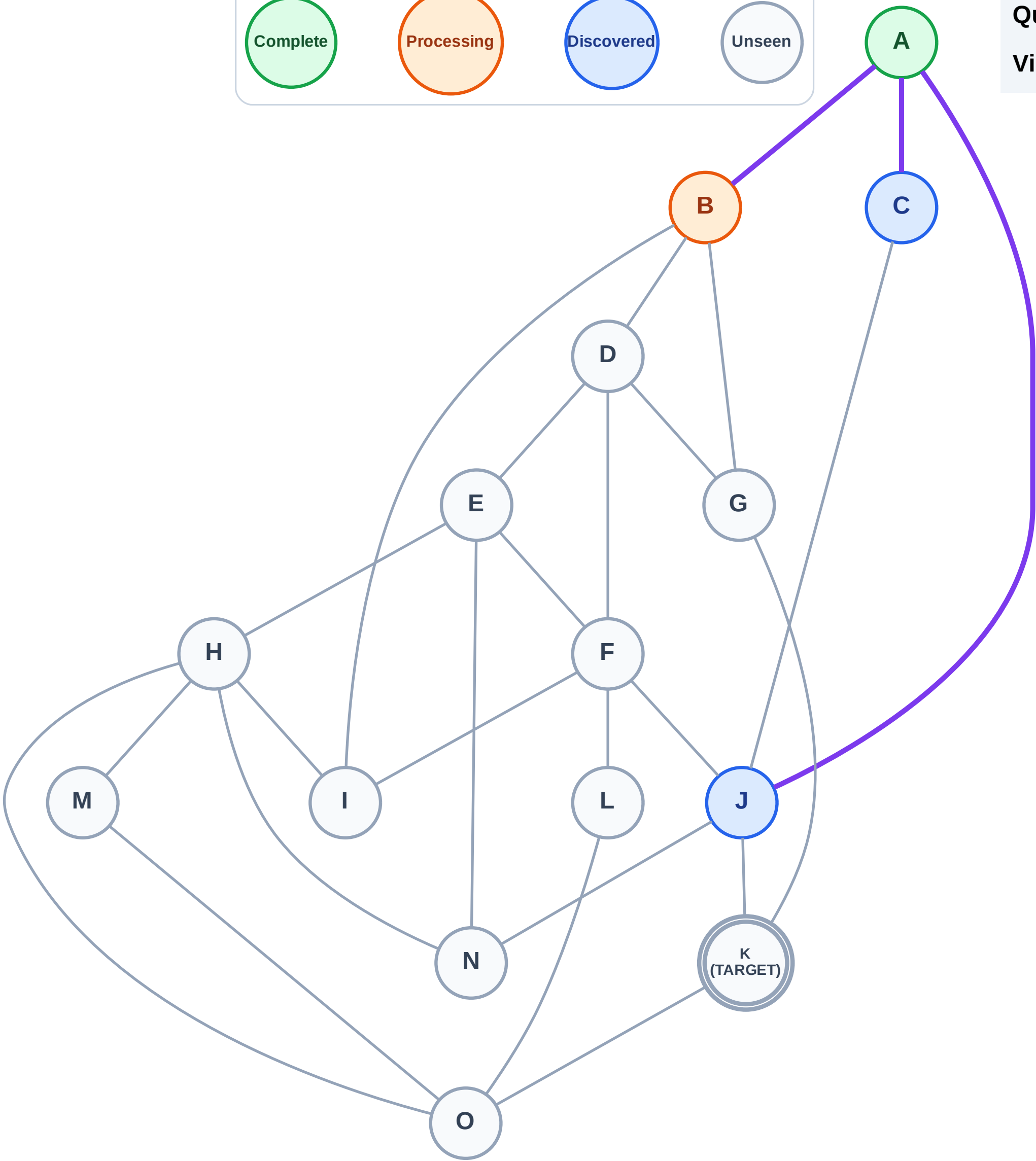
Step 4: Process node B

Legend



Queue: C → J

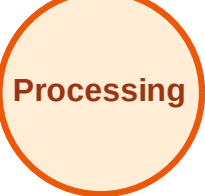
Visited: A



BFS Traversal

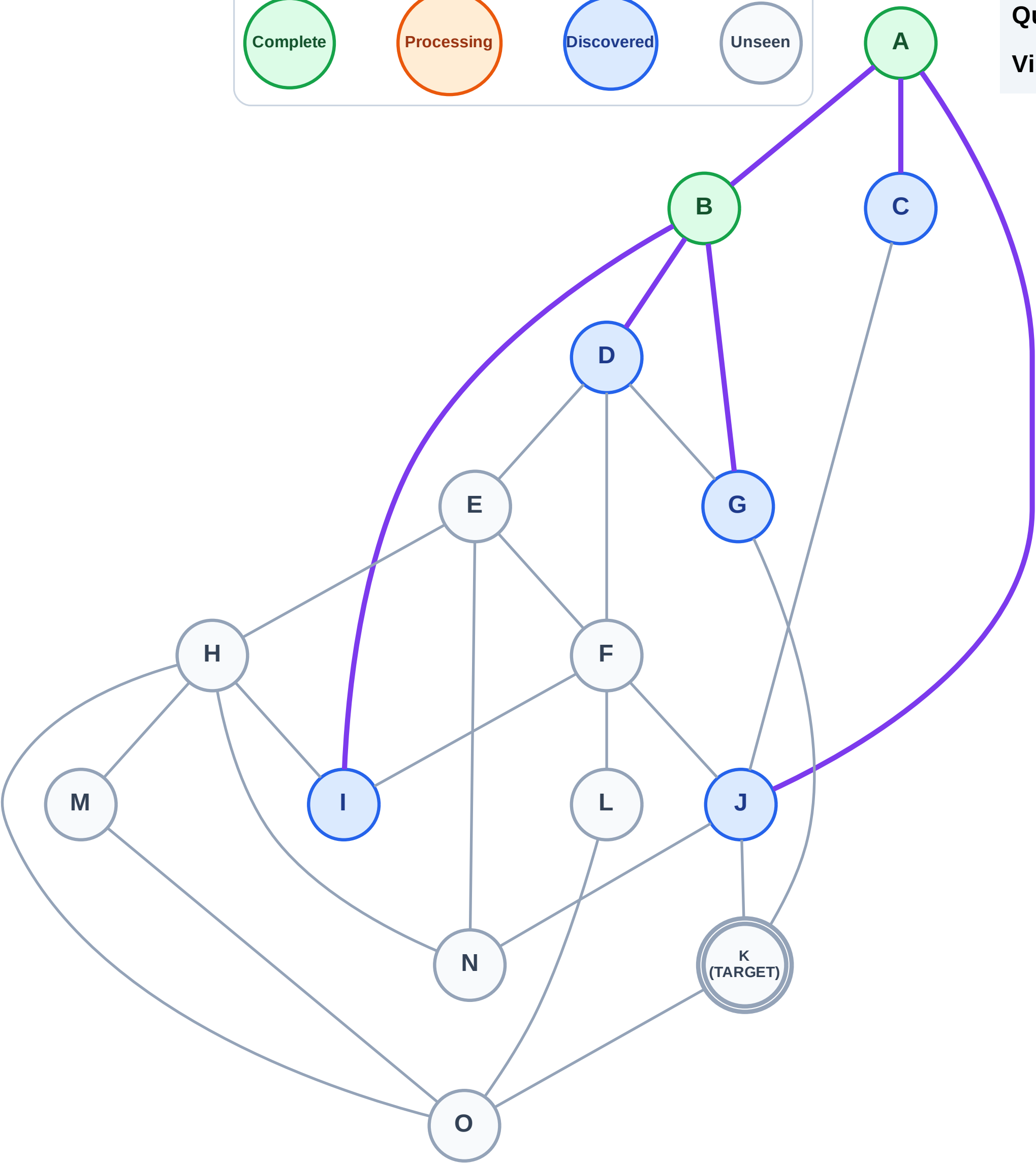
Step 5: Discover D, G, I from B

Legend



Queue: C → J → D → G → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

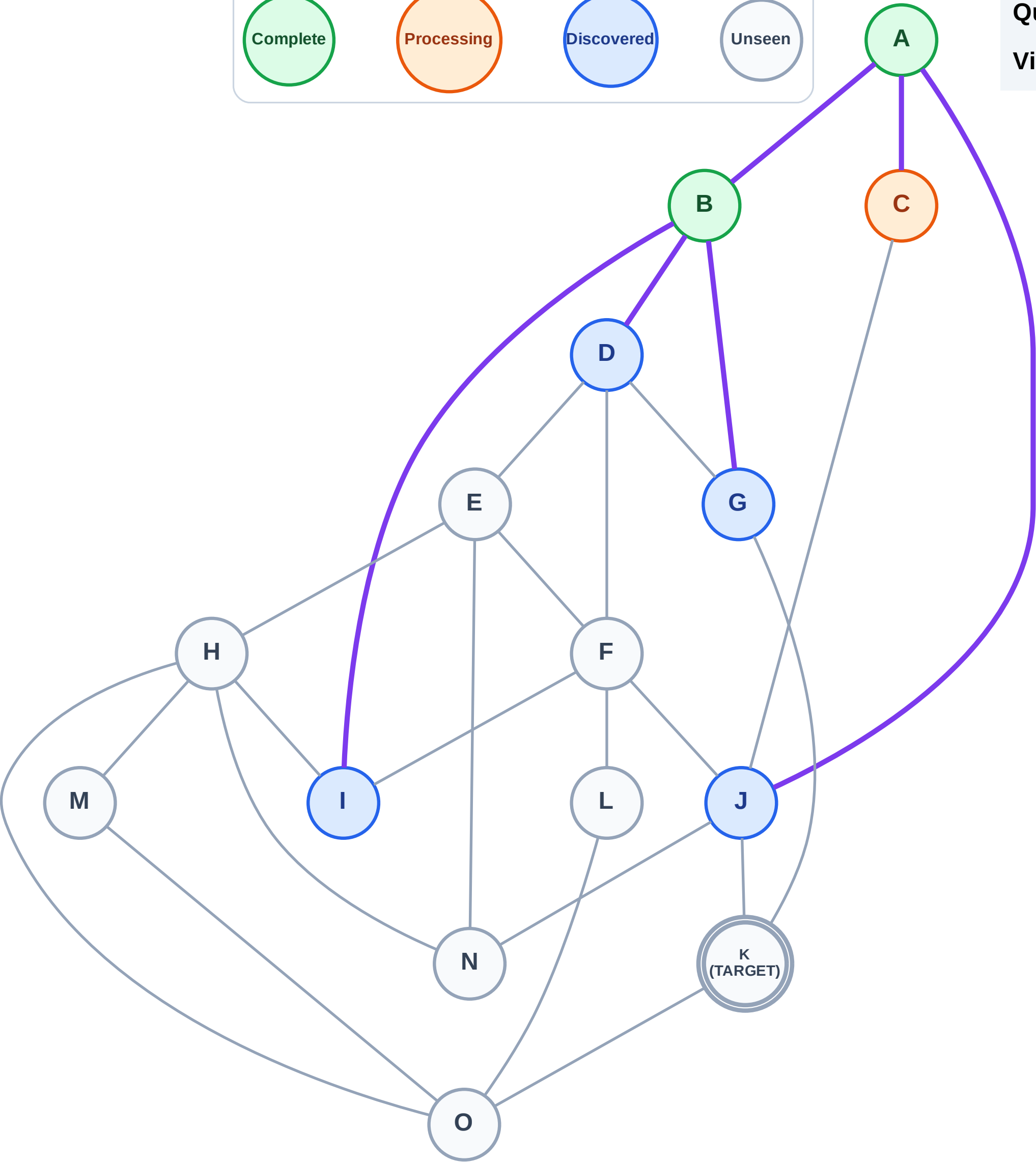
Processing

Discovered

Unseen

Queue: J → D → G → I

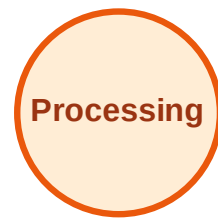
Visited: A, B



BFS Traversal

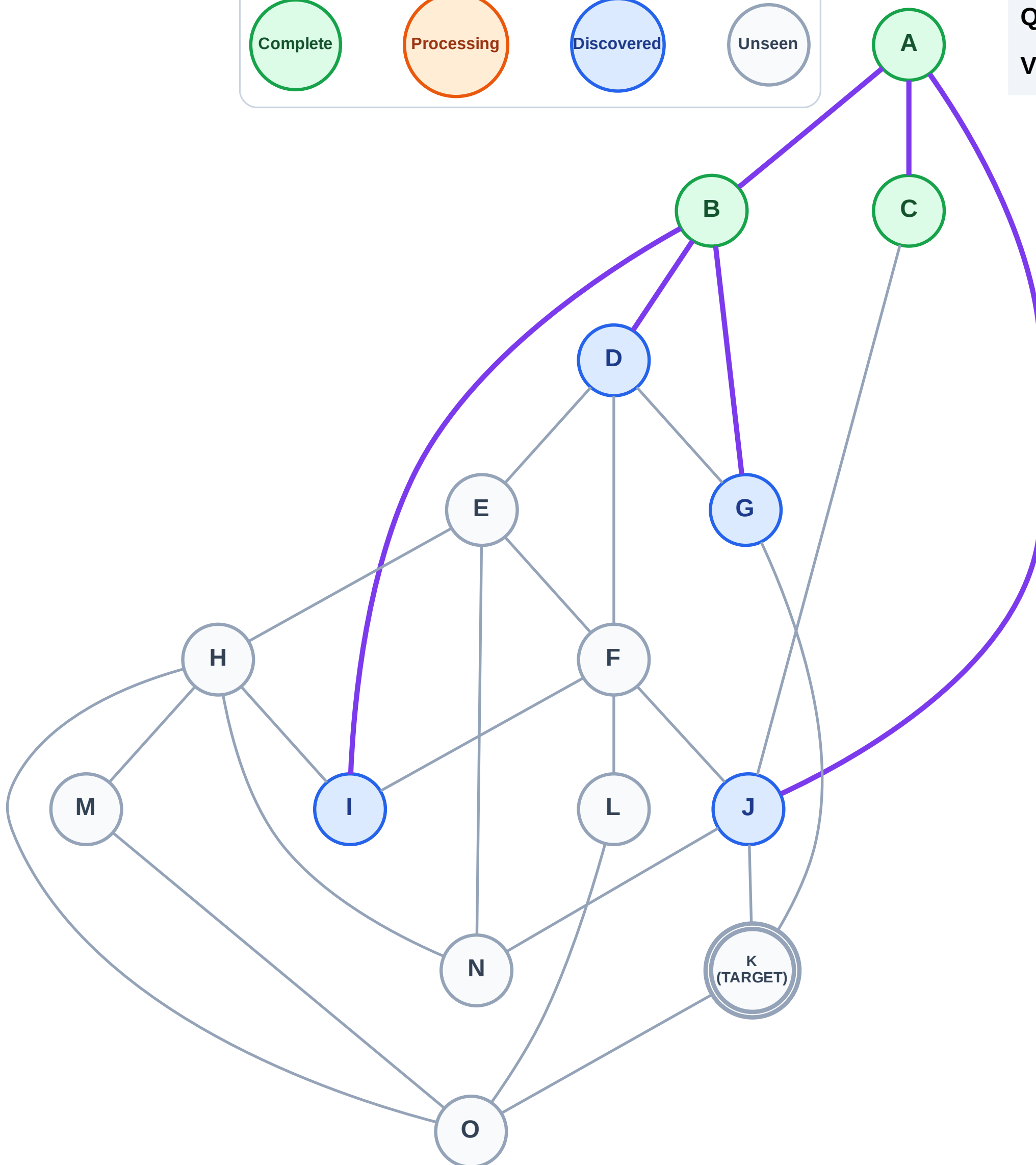
Step 7: No new neighbors from C

Legend



Queue: J → D → G → I

Visited: A, B, C



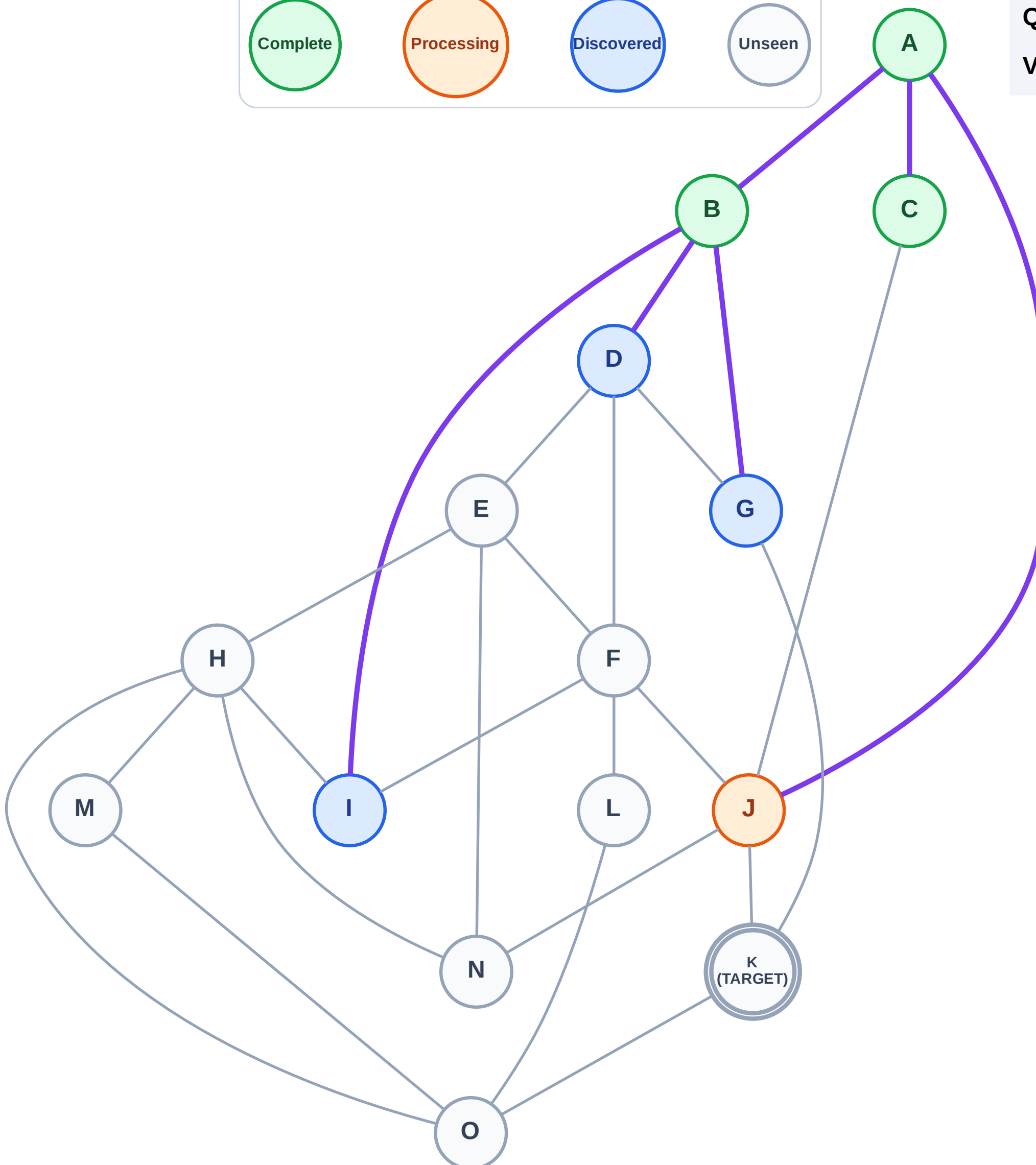
Step 8: Process node J

Step 8: Process node J

Legend



Visited: A, B, C



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → J → K

Legend

Complete

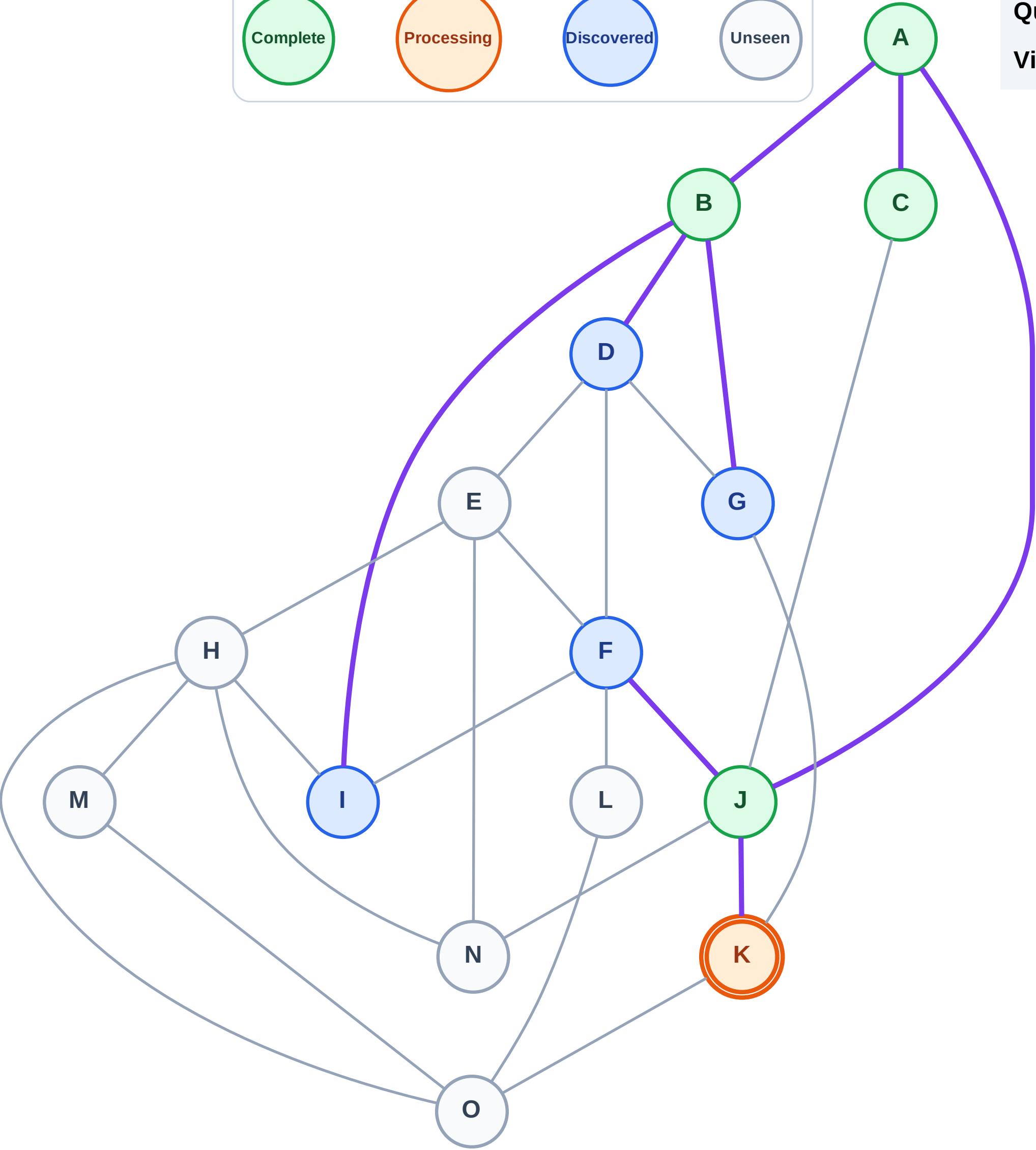
Processing

Discovered

Unseen

Queue: D → G → I → F

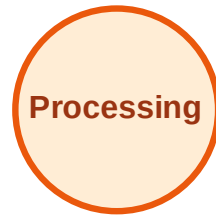
Visited: A, B, C, J, K



DFS Traversal

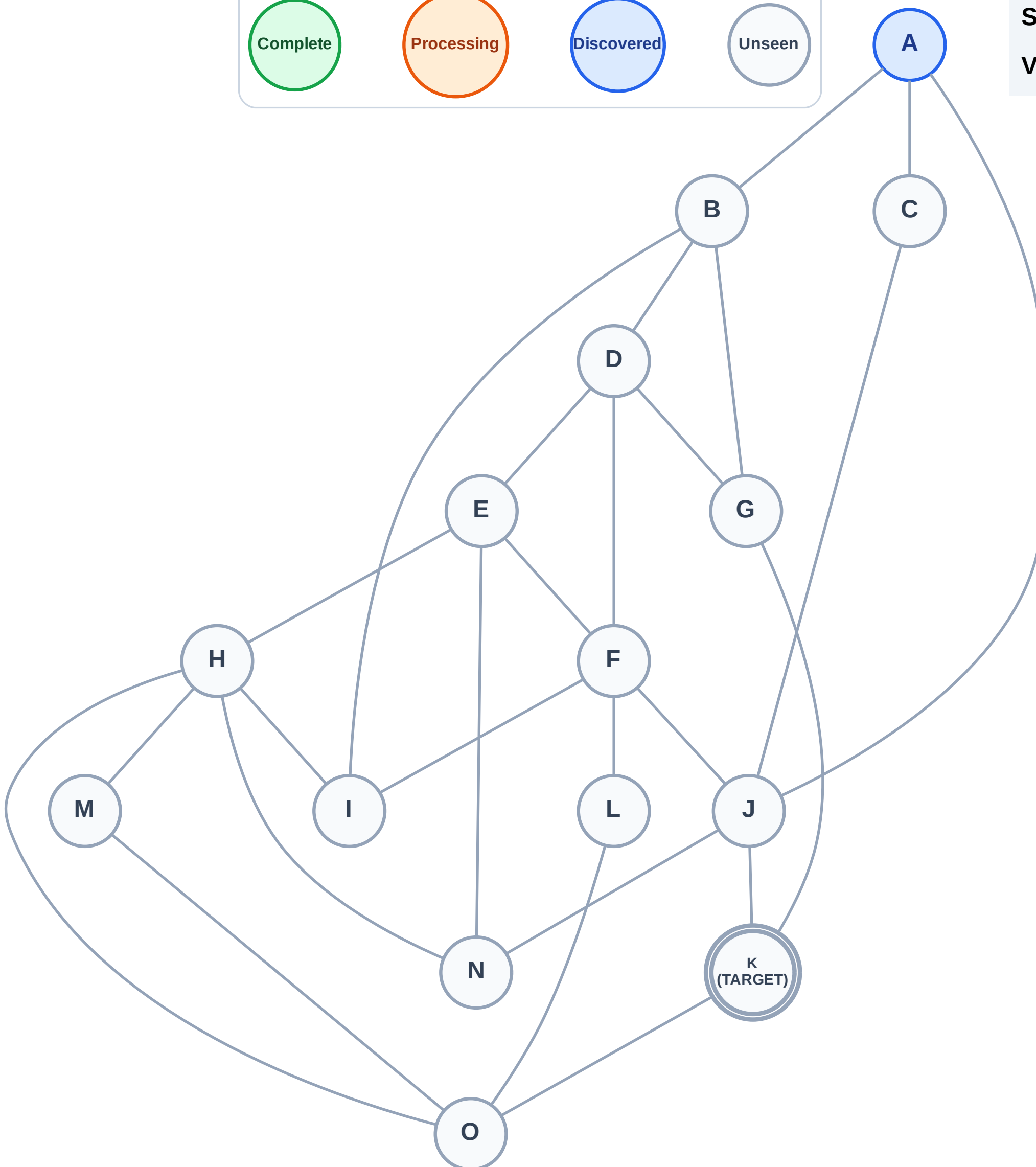
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

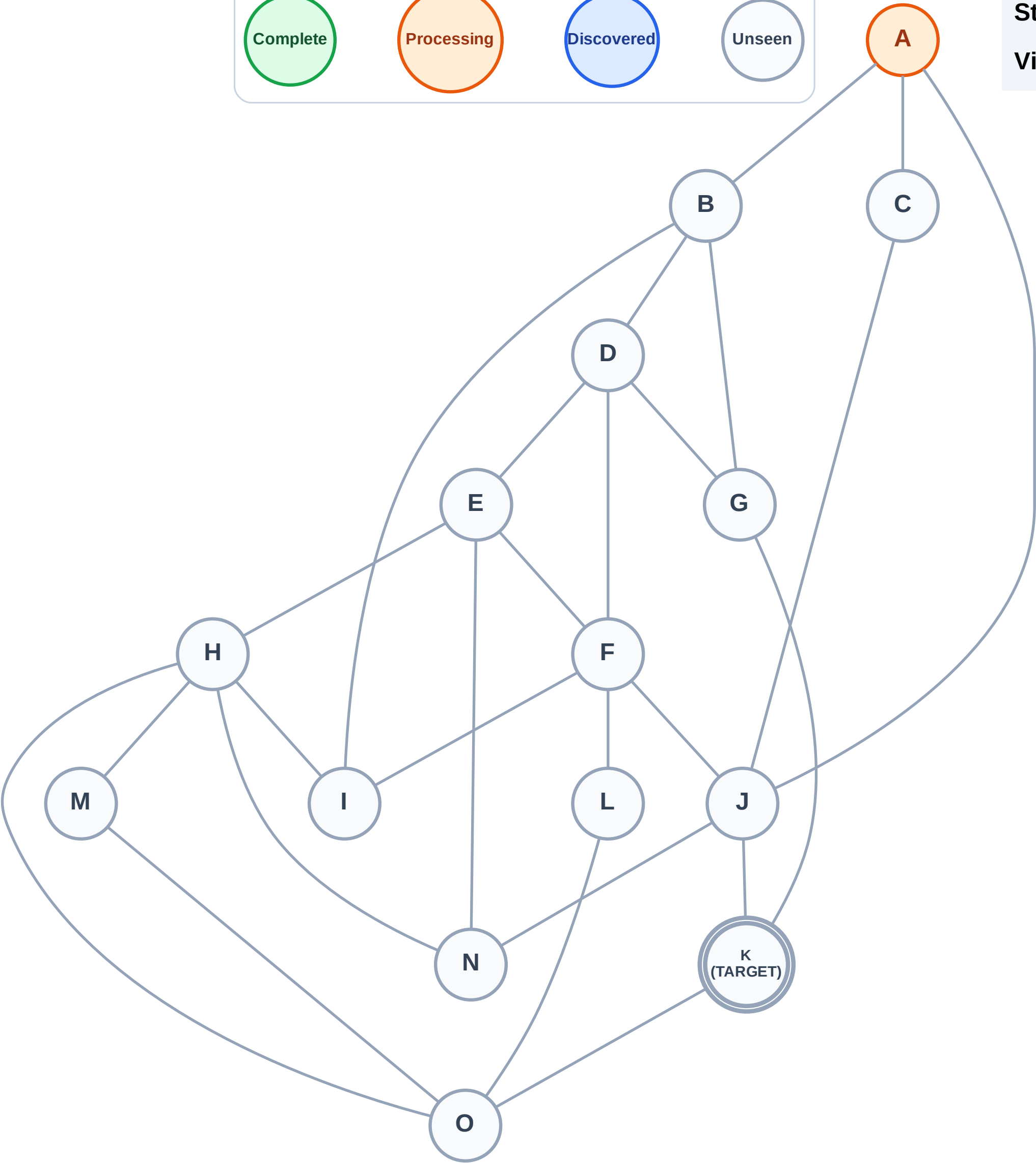
Complete

Processing

Discovered

Unseen

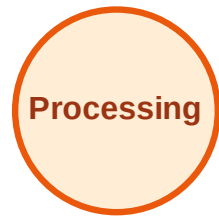
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

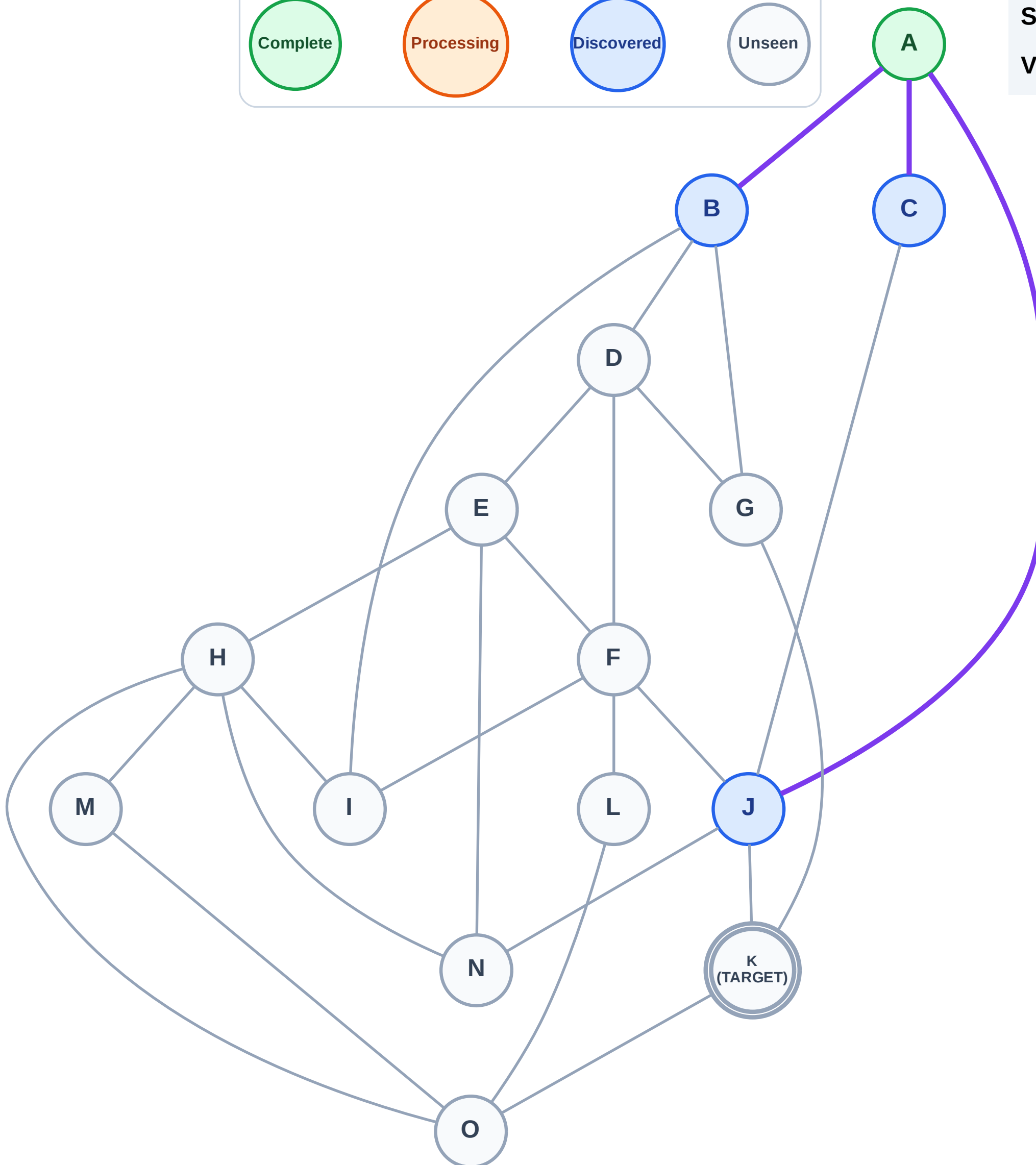
Step 3: Discover J, C, B from A

Legend



Stack (top at right): J | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

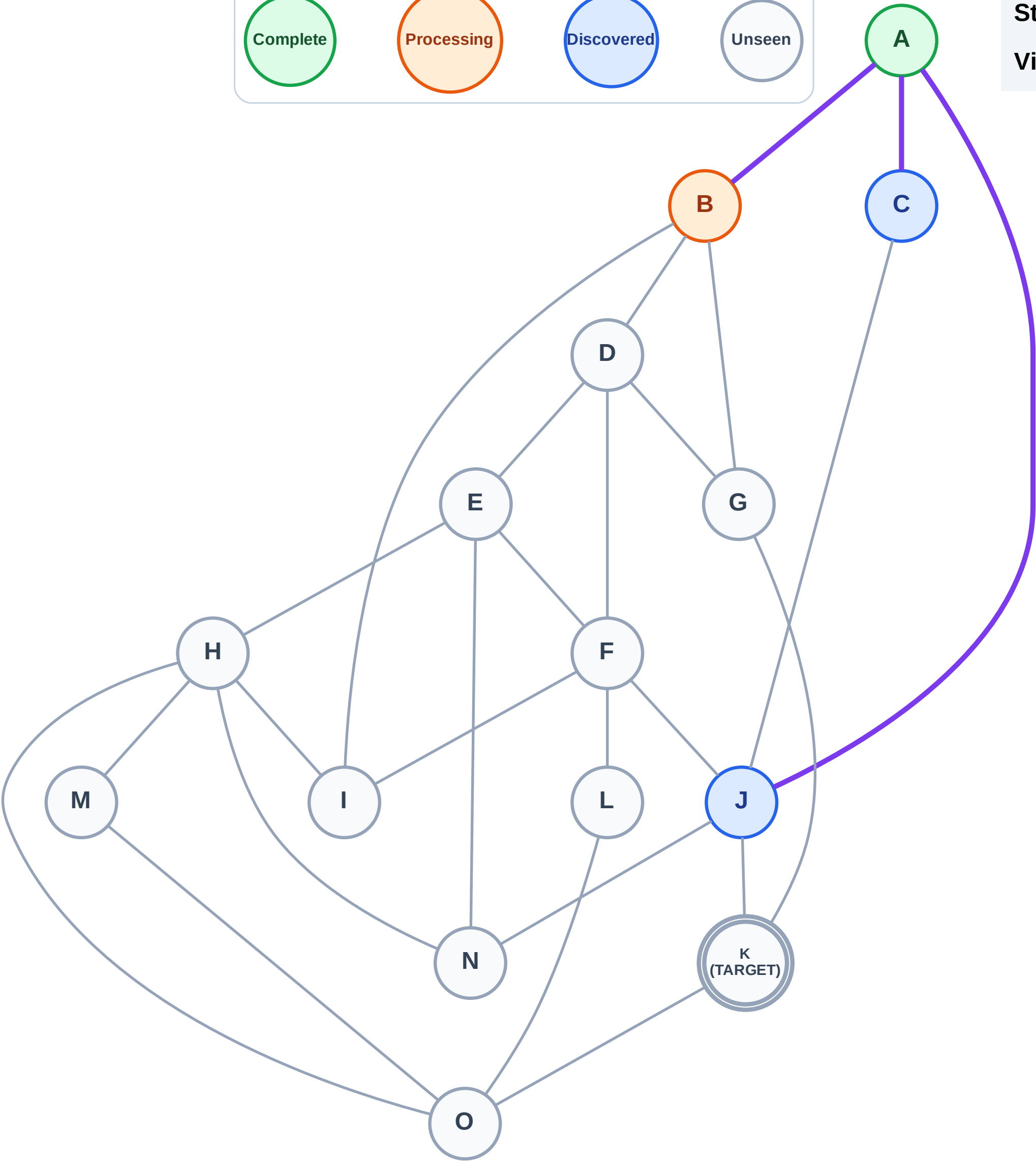
Complete

Processing

Discovered

Unseen

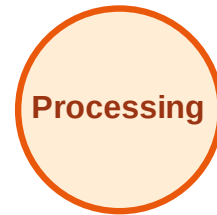
Stack (top at right): J | C
Visited: A



DFS Traversal

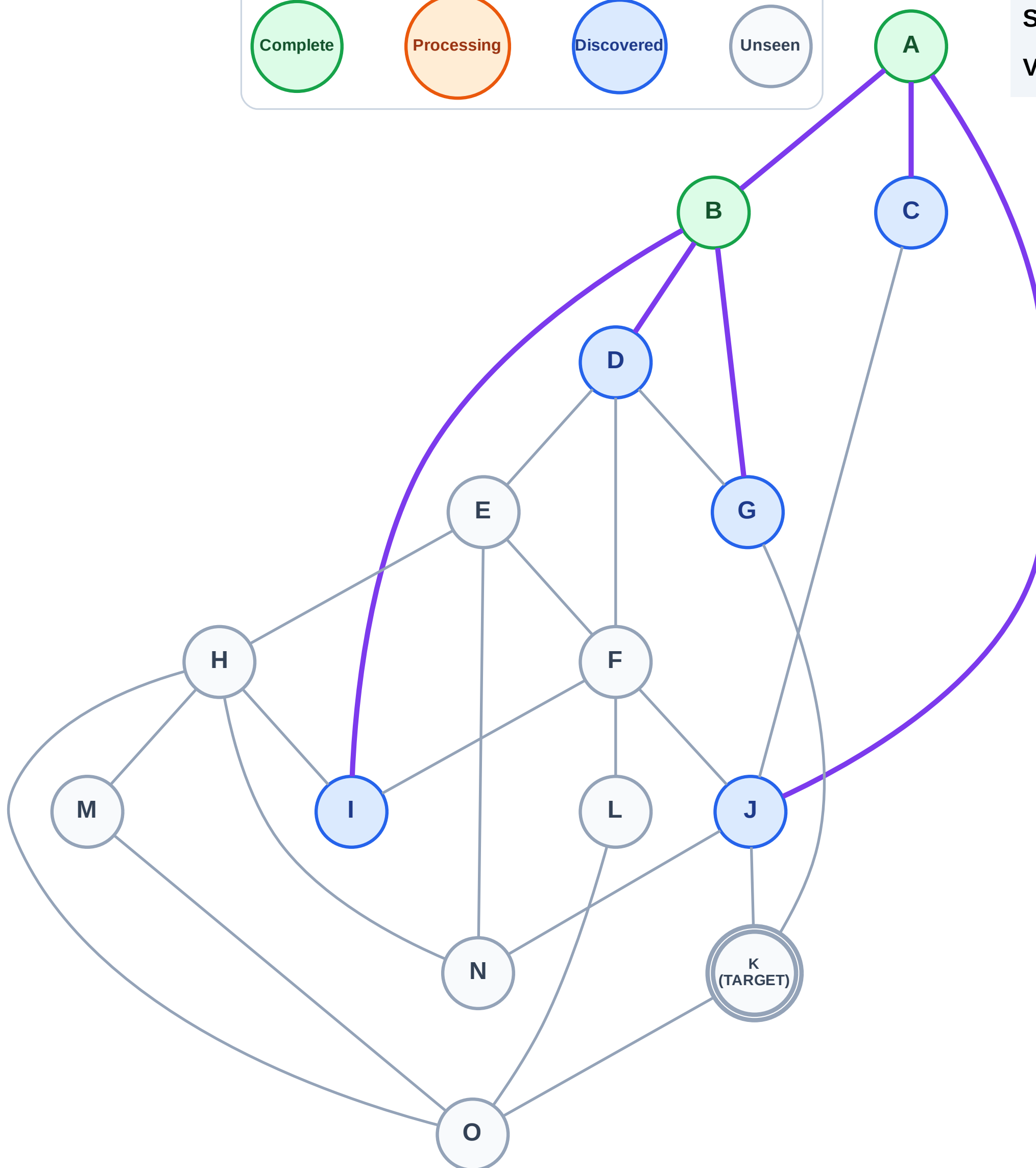
Step 5: Discover I, G, D from B

Legend



Stack (top at right): J | C | I | G | D

Visited: A, B



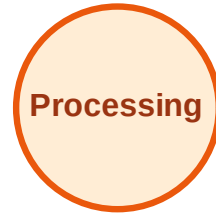
DFS Traversal

Step 6: Process node D

Legend



Complete



Processing



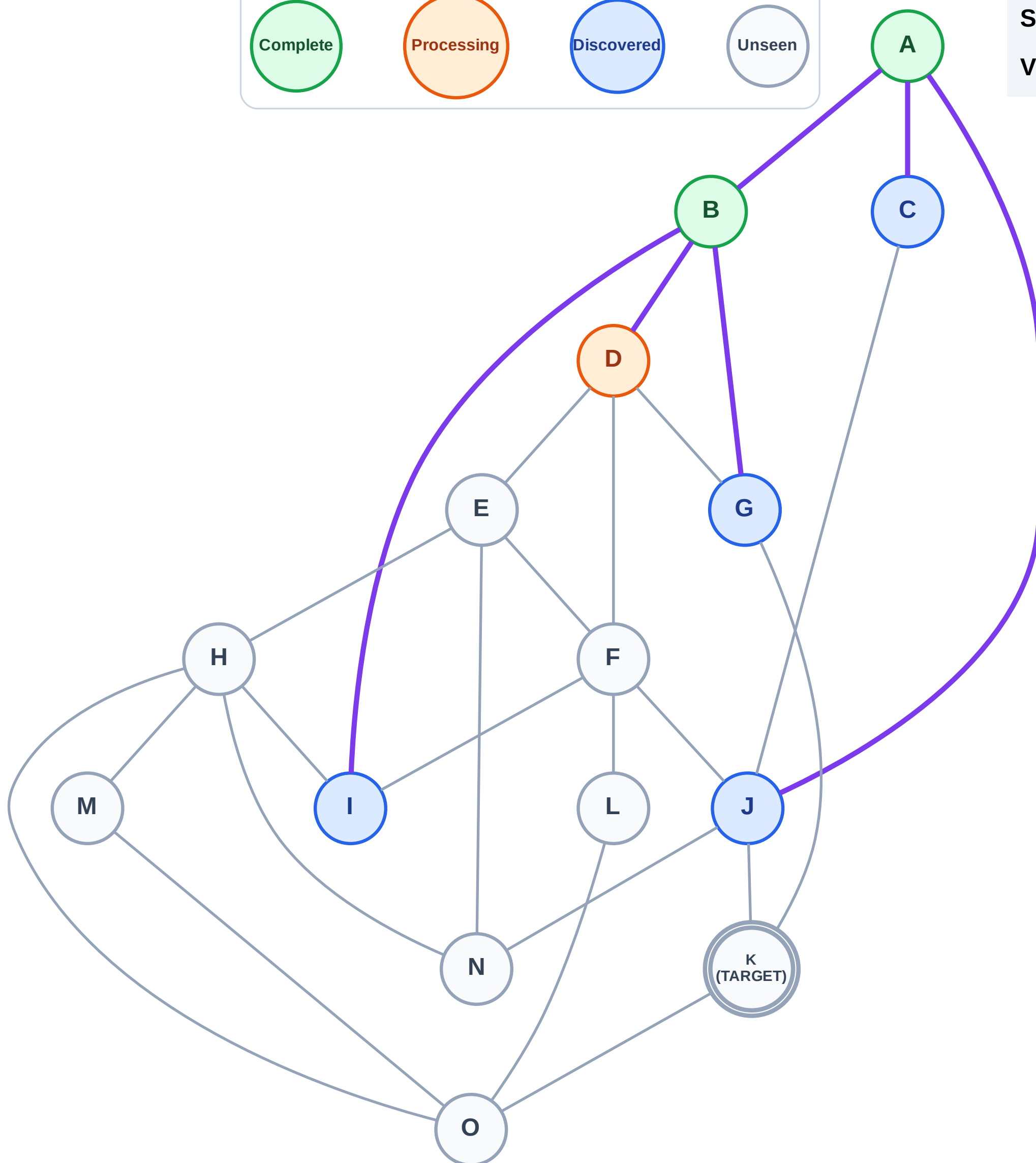
Discovered



Unseen

Stack (top at right): J | C | I | G

Visited: A, B



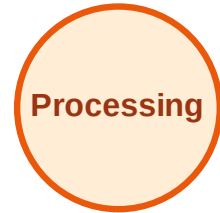
DFS Traversal

Step 7: Discover F, E from D

Legend



Complete



Processing



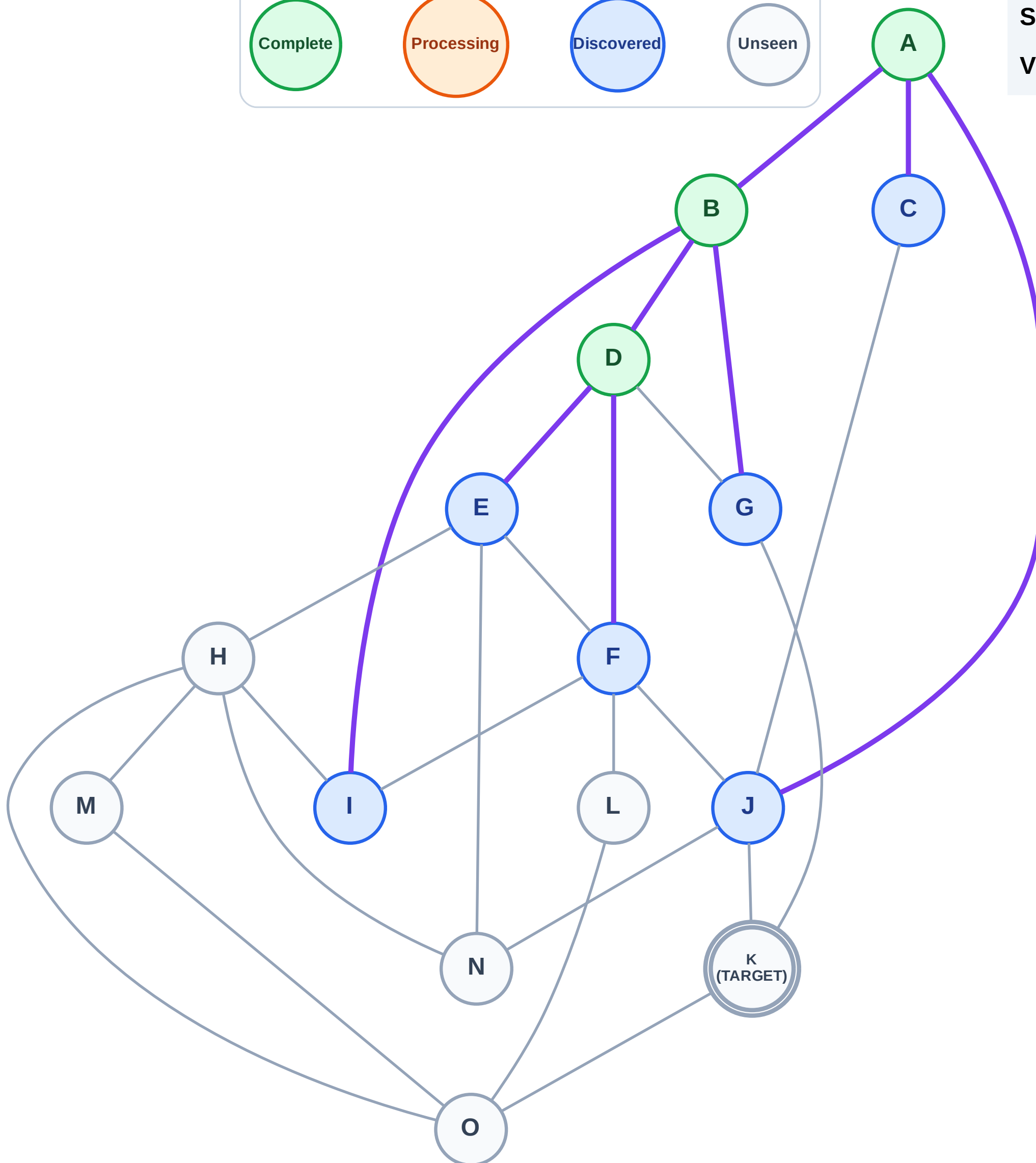
Discovered



Unseen

Stack (top at right): J | C | I | G | F | E

Visited: A, B, D



DFS Traversal

Step 8: Process node E

Legend

Complete

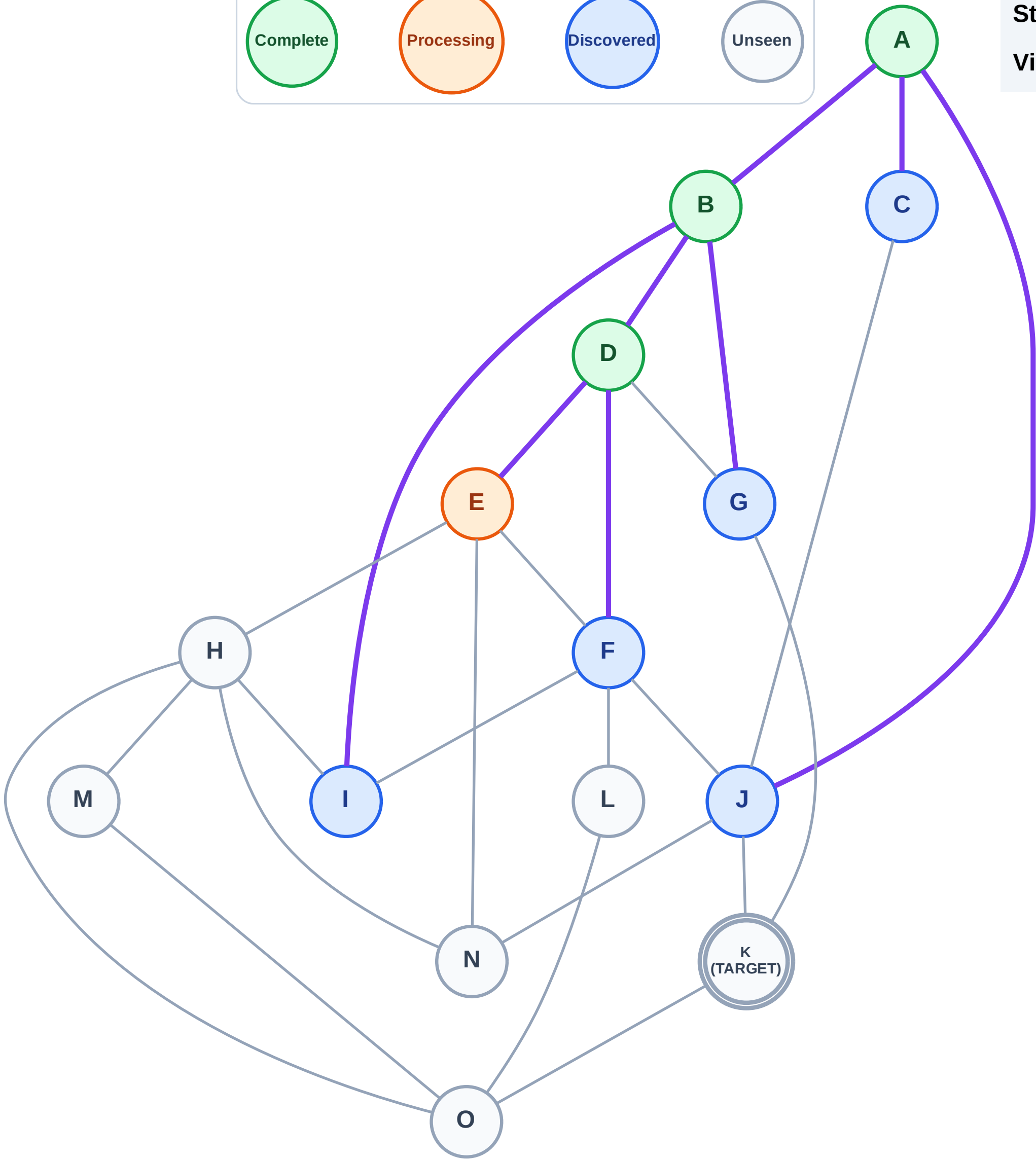
Processing

Discovered

Unseen

Stack (top at right): J | C | I | G | F

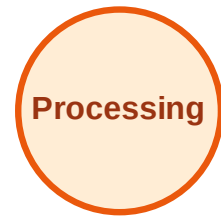
Visited: A, B, D



DFS Traversal

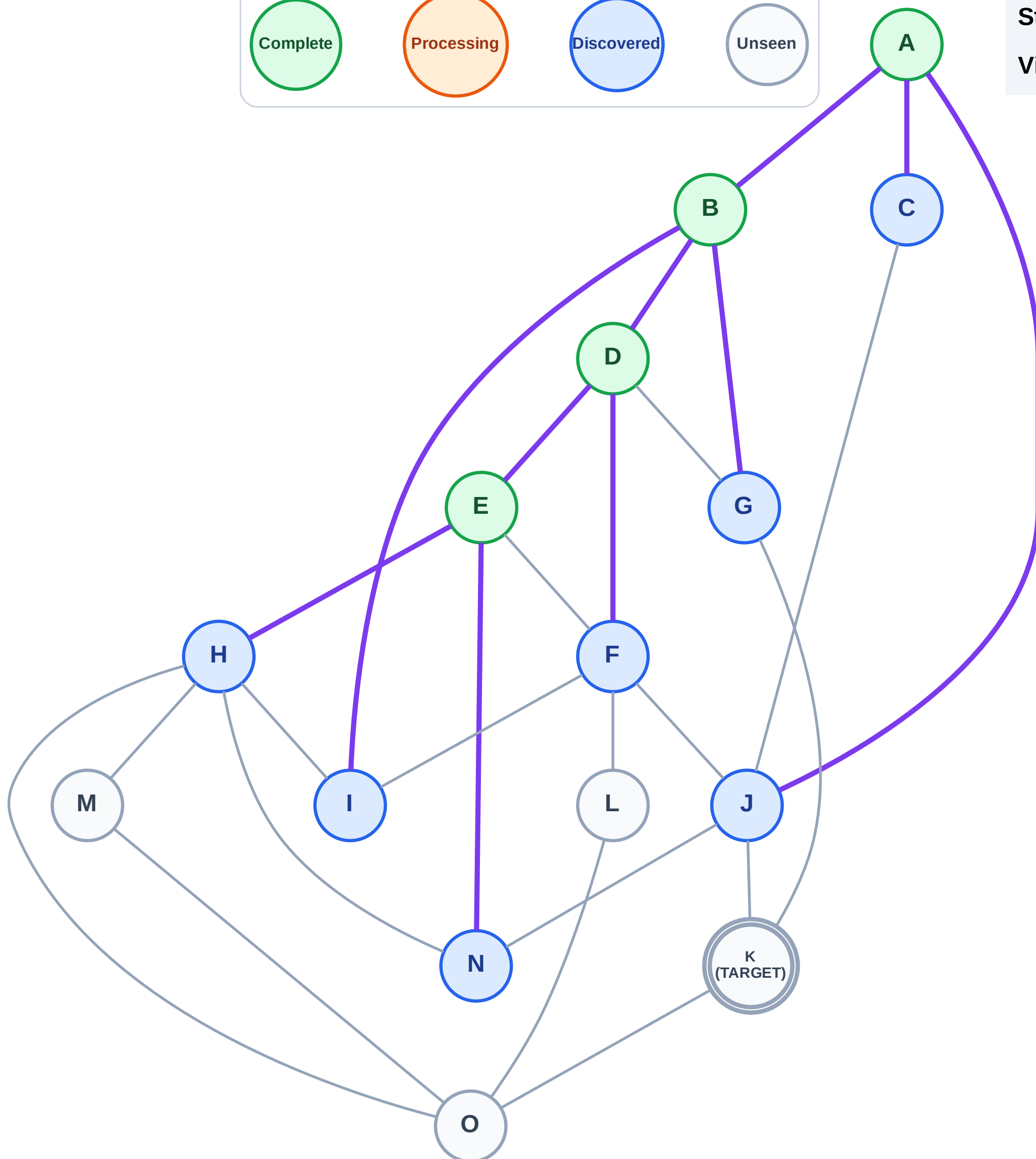
Step 9: Discover N, H from E

Legend



Stack (top at right): J | C | I | G | F | N | H

Visited: A, B, D, E



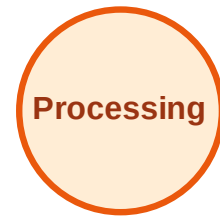
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



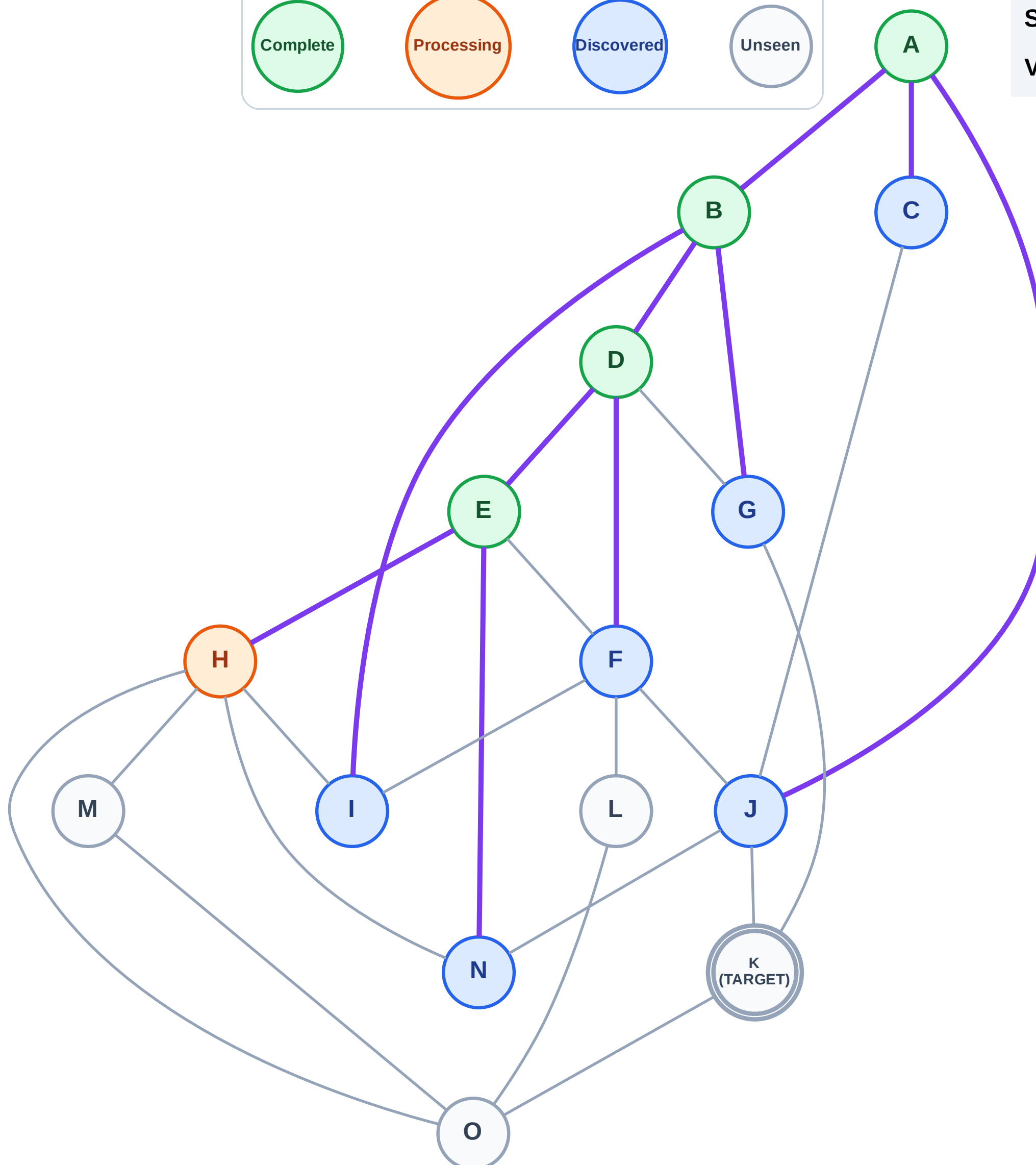
Discovered



Unseen

Stack (top at right): J | C | I | G | F | N

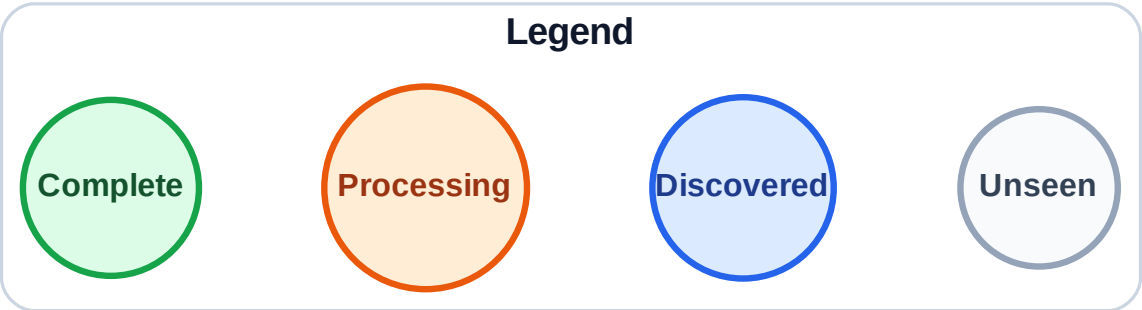
Visited: A, B, D, E



DFS Traversal

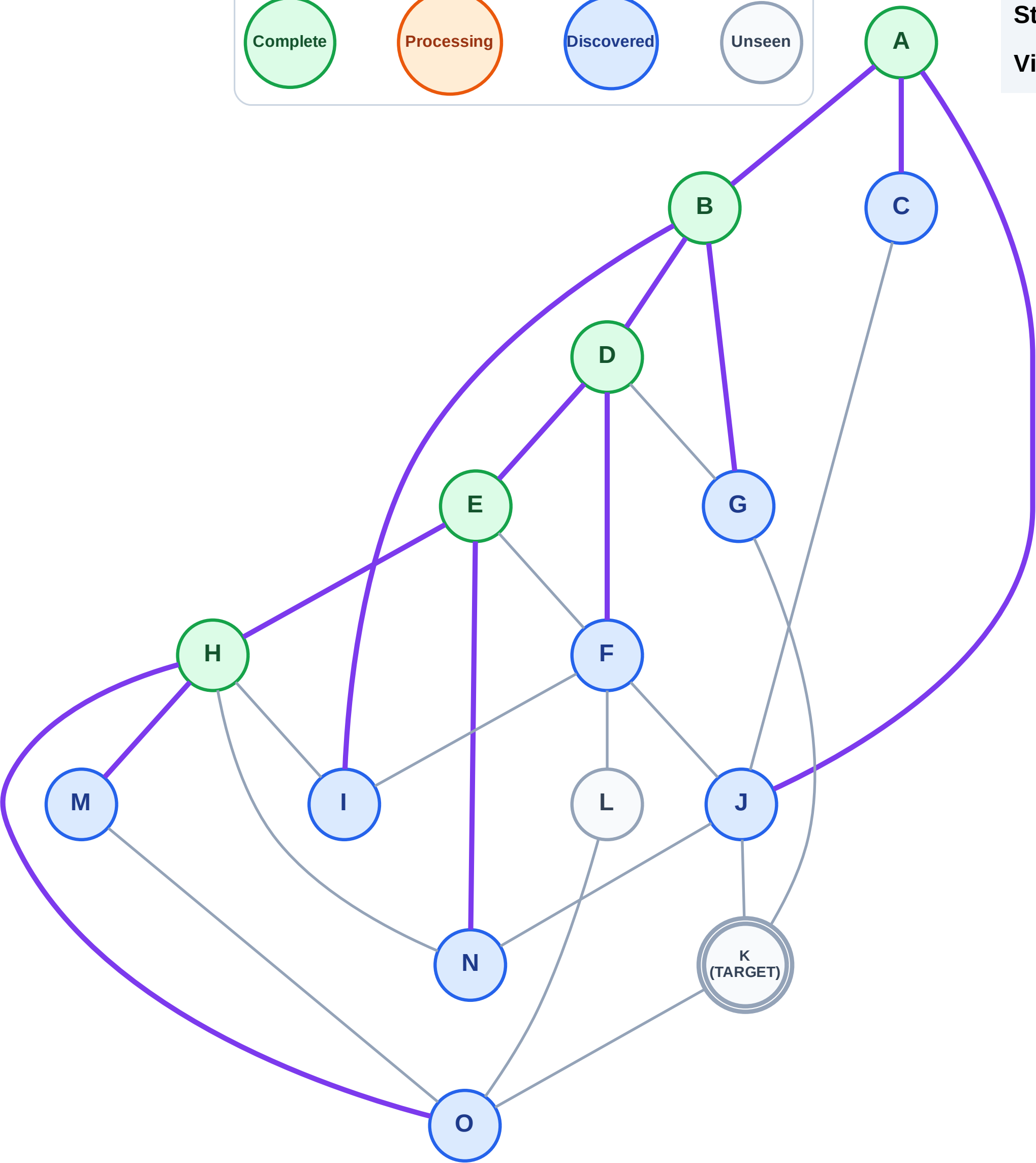
Step 11: Discover O, M from H

Legend



Stack (top at right): J | C | I | G | F | N | O | M

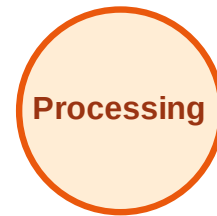
Visited: A, B, D, E, H



DFS Traversal

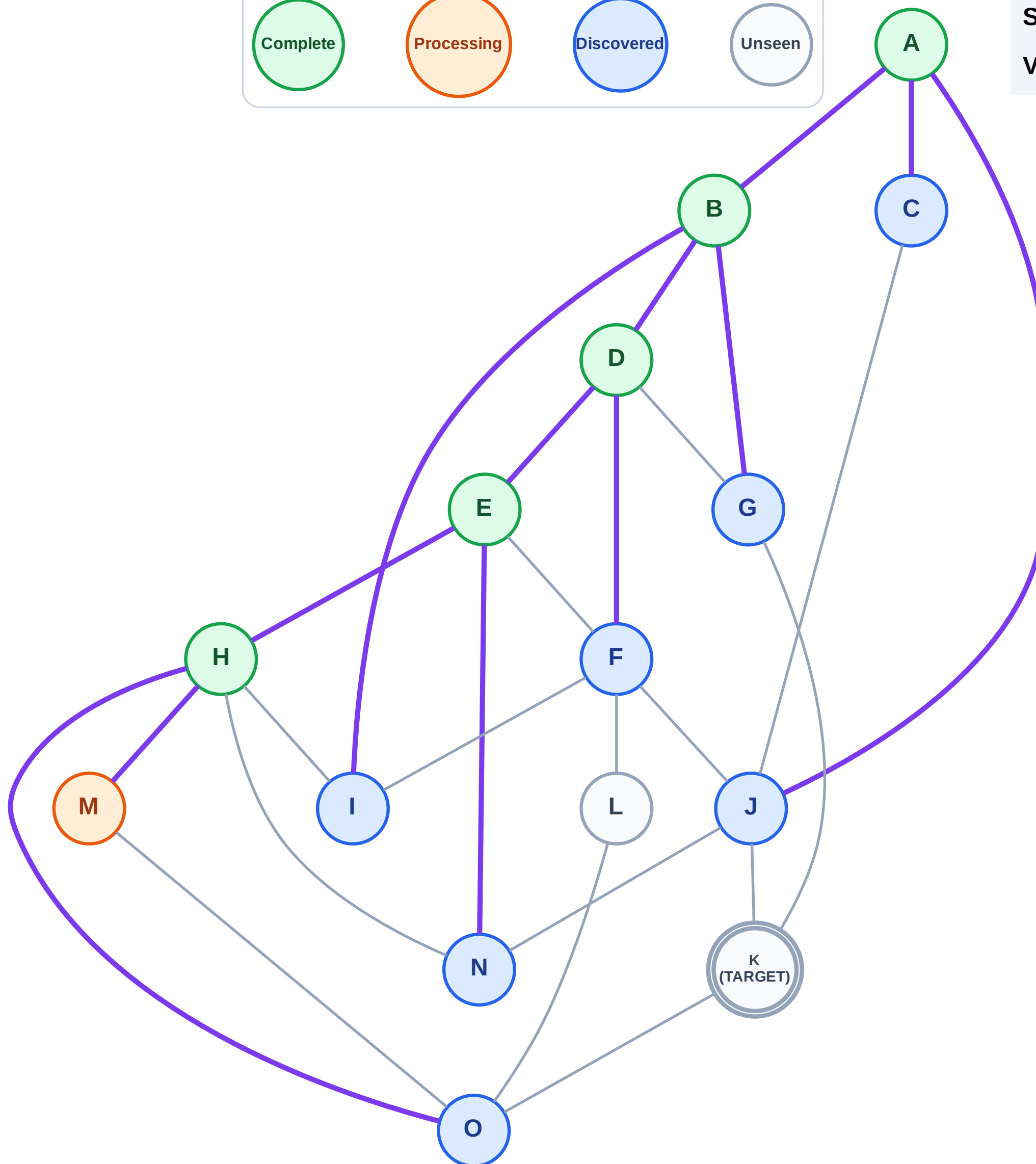
Step 12: Process node M

Legend



Stack (top at right): J | C | I | G | F | N | O

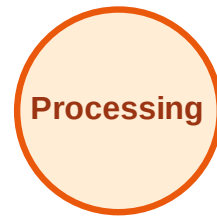
Visited: A, B, D, E, H



DFS Traversal

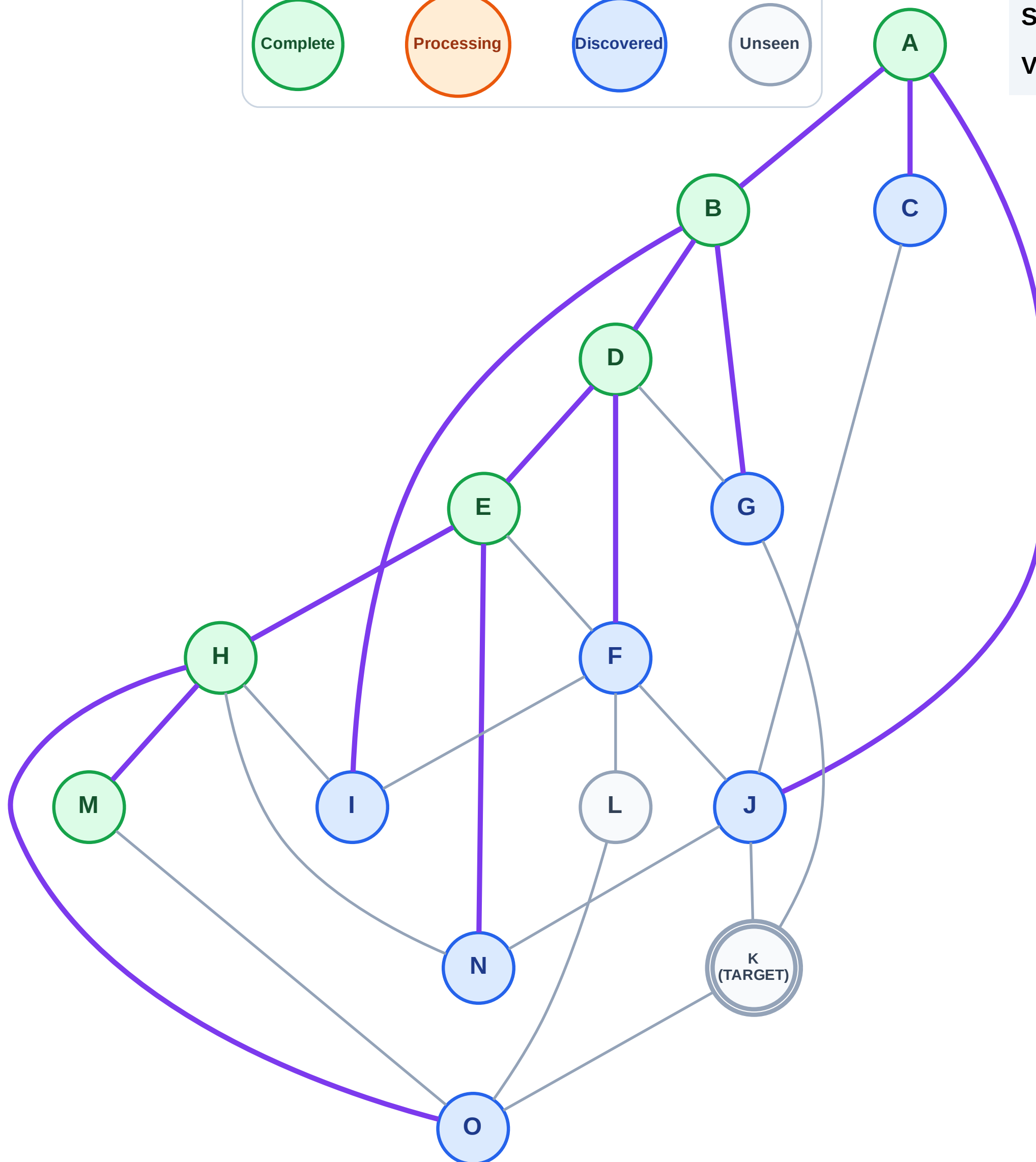
Step 13: No new neighbors from M

Legend



Stack (top at right): J | C | I | G | F | N | O

Visited: A, B, D, E, H, M



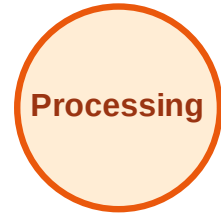
DFS Traversal

Step 14: Process node O

Legend



Complete



Processing



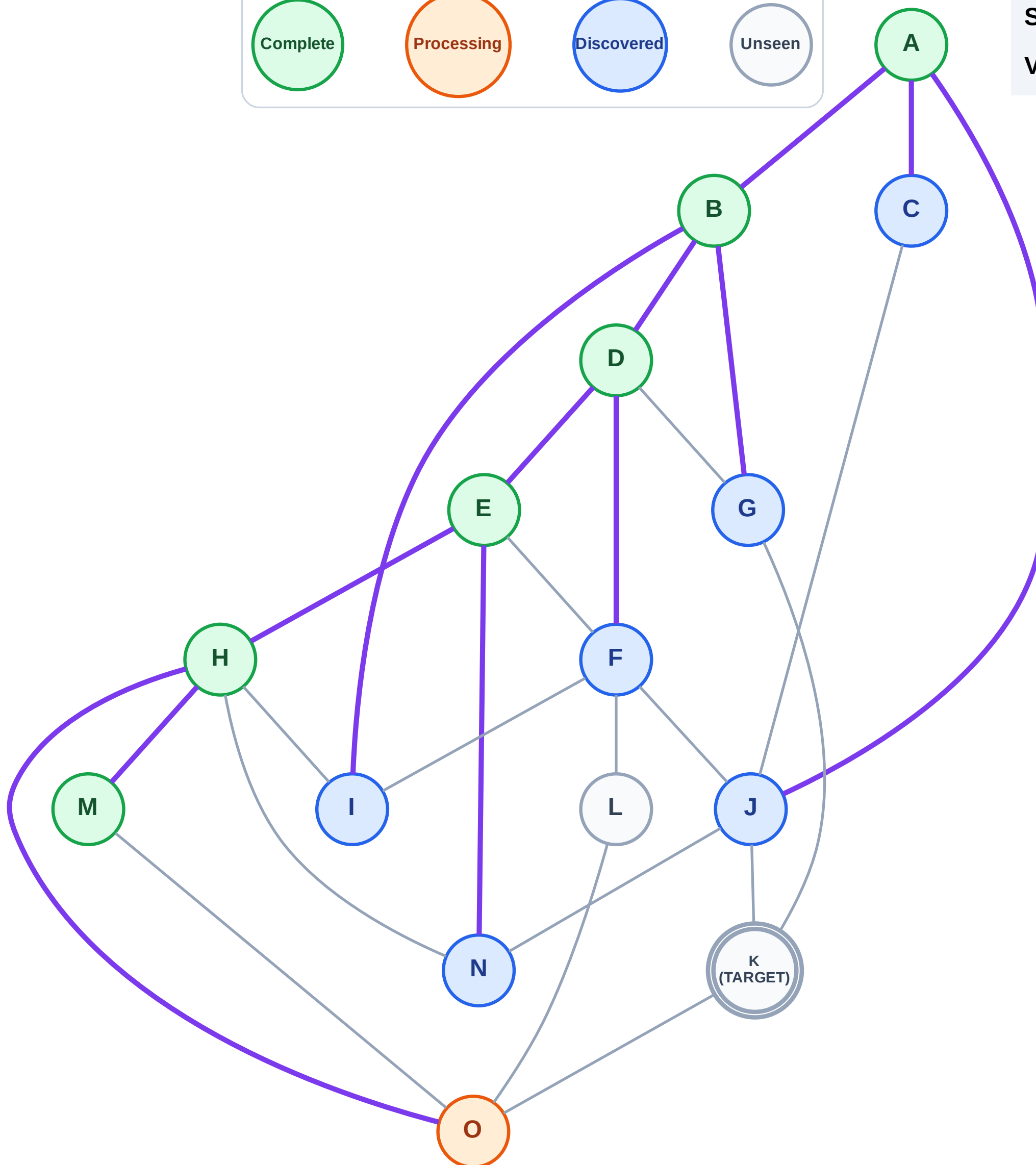
Discovered



Unseen

Stack (top at right): J | C | I | G | F | N

Visited: A, B, D, E, H, M



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (6 hops): A → B → D → E → H → O → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): J | C | I | G | F | N | L

Visited: A, B, D, E, H, K, M, O

